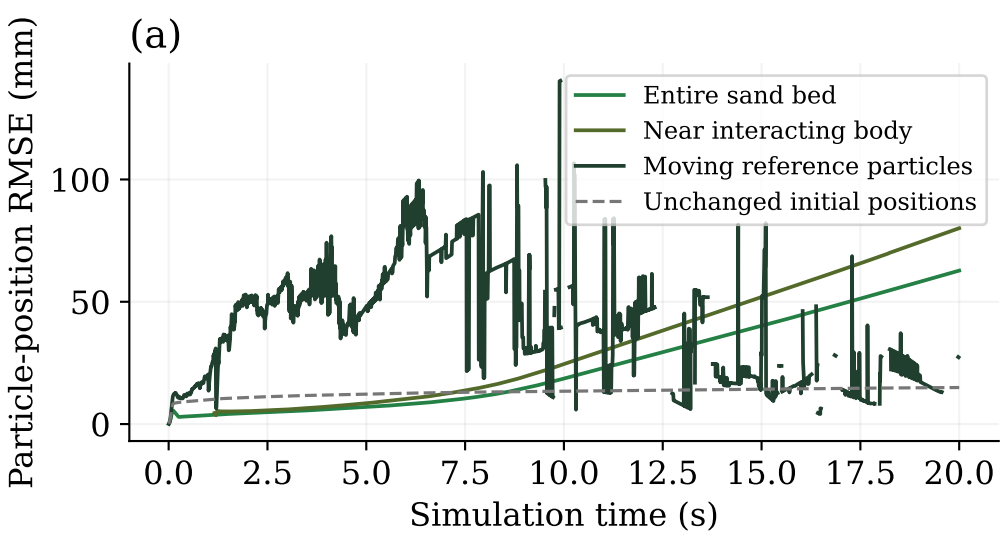


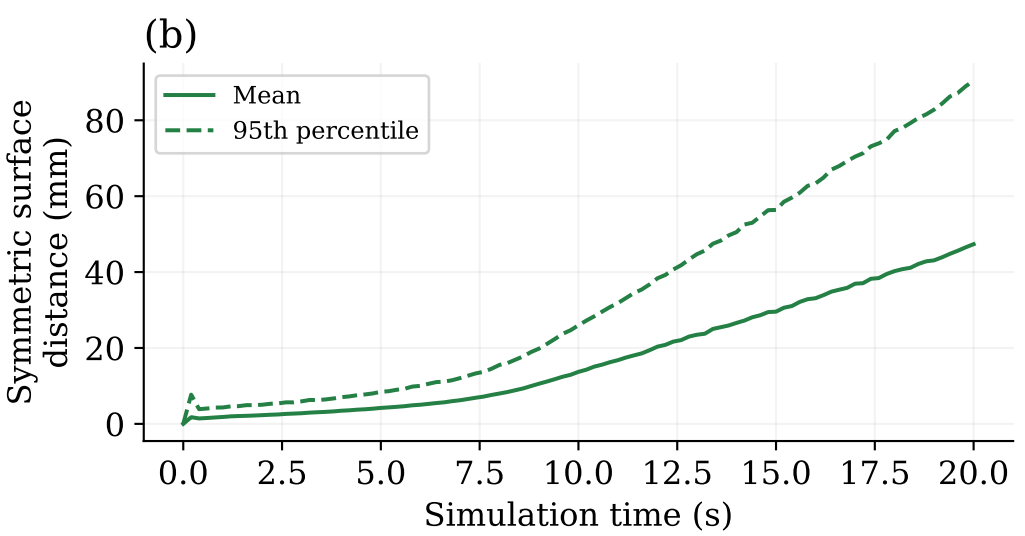


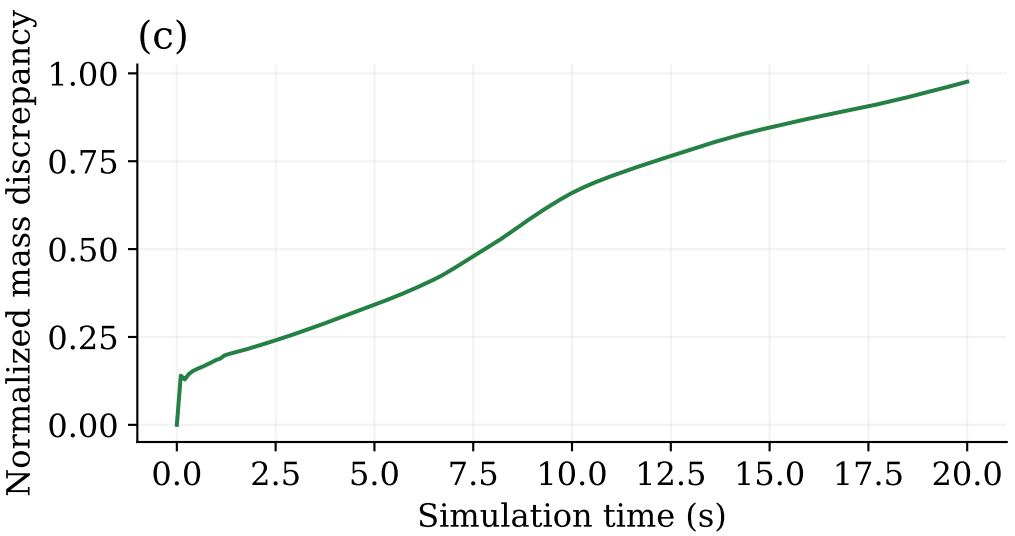


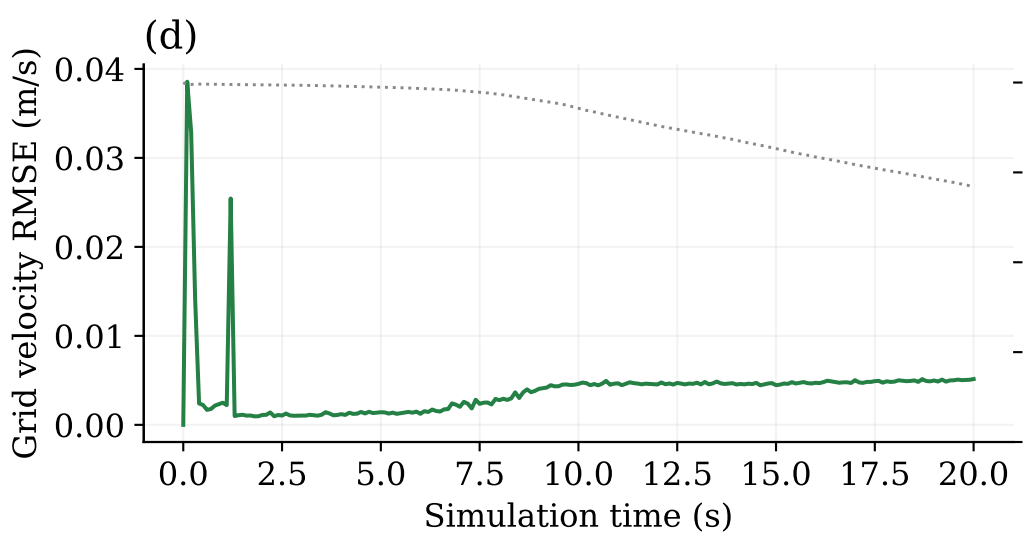


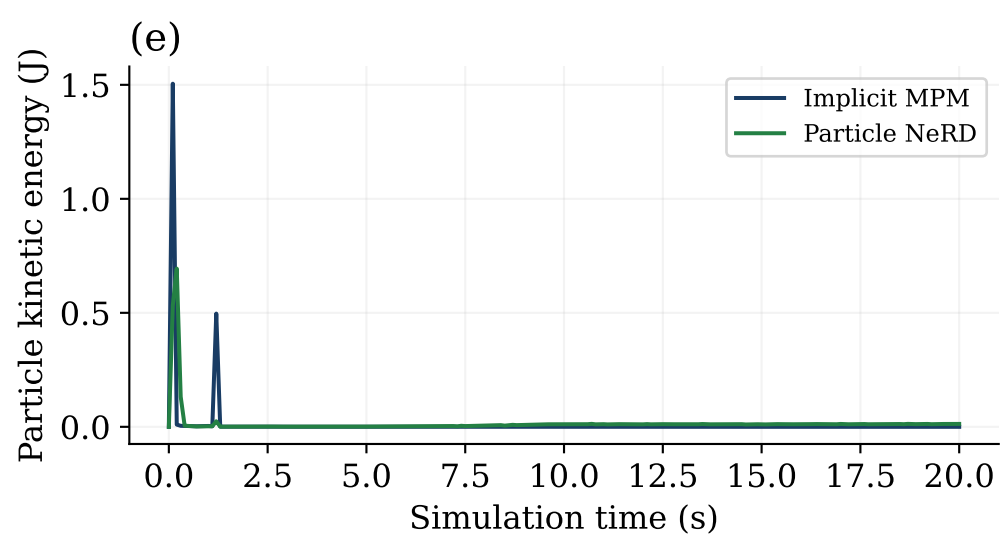


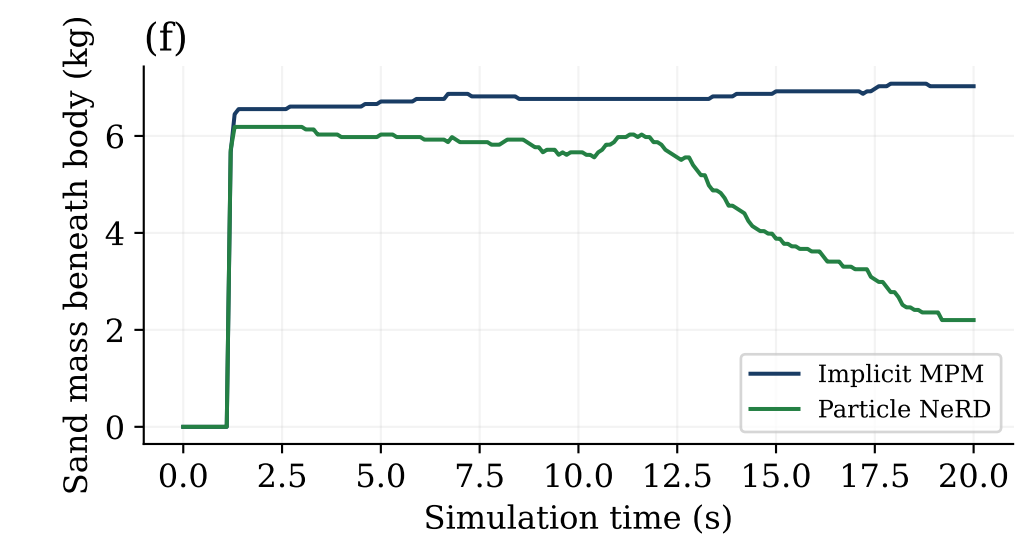


Figure 1: One paired rollout with identical initial particles and recorded body motion.
(a) Corresponding-particle position error, including a whole-bed unchanged-initial-position baseline; (b) sand-surface error;
(c) physical-mass redistribution error; (d) material velocity-field error; (e) particle kinetic energy;
(f) sand mass in the body's horizontal bounding rectangle, within 0.10 m below its lowest point and above the floor.
Near-contact reference distance ≤0.05 m; moving reference speed >0.01 m/s. Empty groups produce gaps; Figure 6 shows counts.
(c) Sum of absolute grid-mass differences divided by total reference mass (dimensionless).
(d) Reference-mass-weighted velocity RMSE in cells occupied by both methods; dotted curve is their
fraction of total reference mass. (b) The surface percentile is over sampled distances, not repeated trials.
The model advances its own particle history at 200 Hz. No displacement gate, particle projection, or force filter.